

	is eliminated.	
7	C Tension in string is $3.0W$ ($2.0W + 1.0W$) Let distance between string and CG be d Taking moments abouts the CG (which is at $0.5x$) $(2.0W) \times (0.3x) = (3.0W) \times d$ $d = 0.2x$ When load is moved, it will be $0.3x$ away from the CG (but on the other side) Hence, the string needs to be placed $0.2x$ from the CG (but on the other side) for rod to be in equilibrium. Distance moved by string = $0.2x + 0.2x = 0.4x$	A
8	A	A